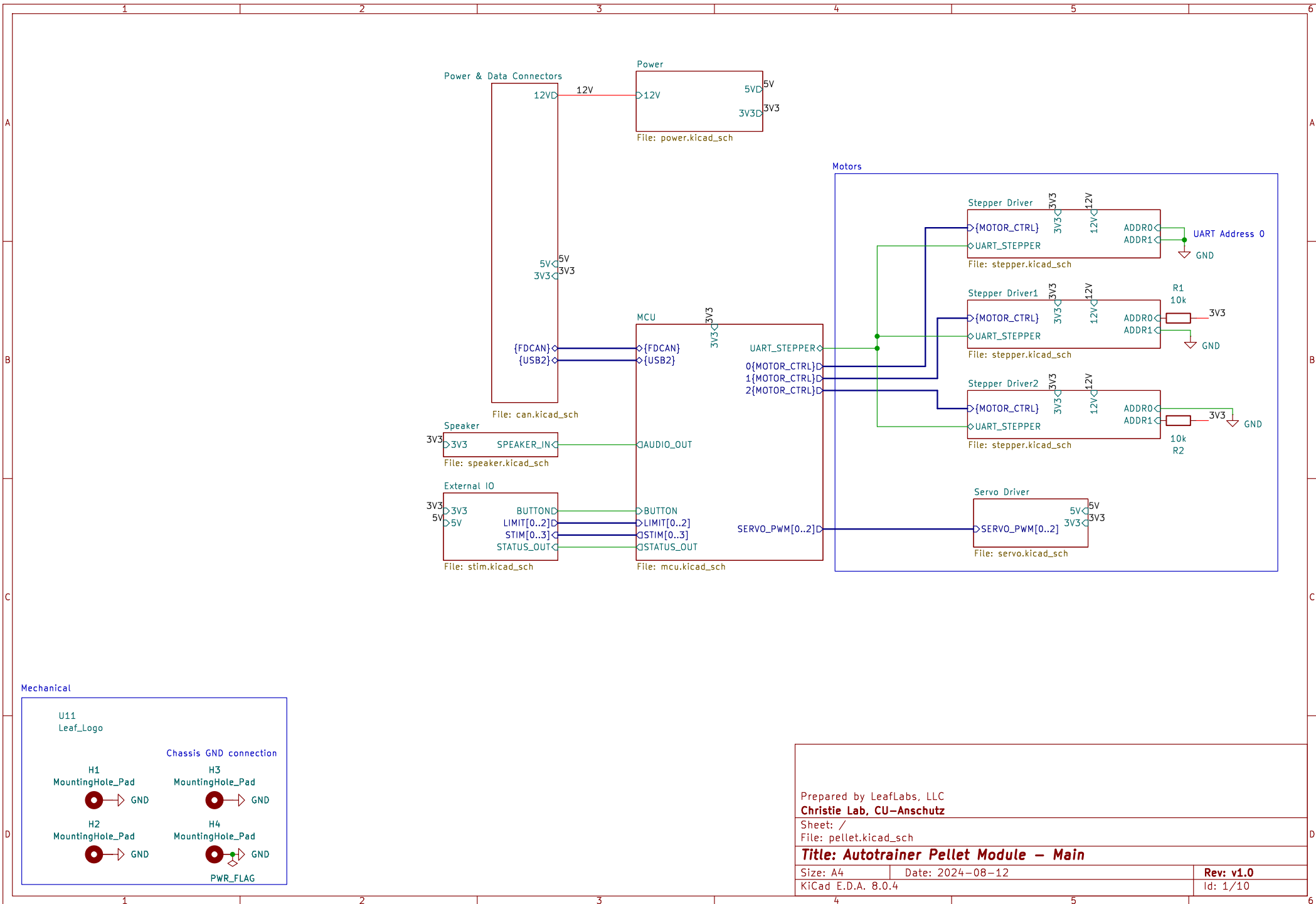
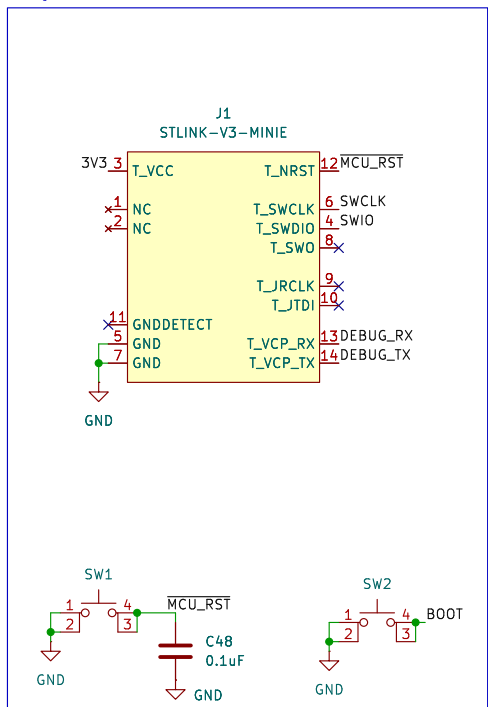
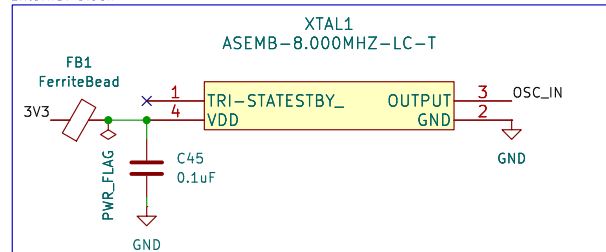




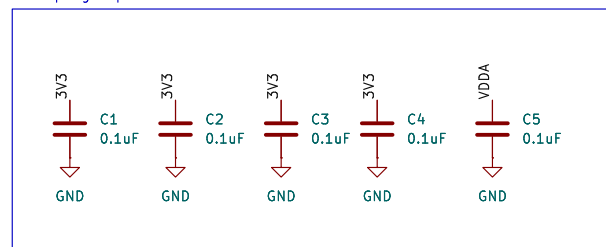
Debug



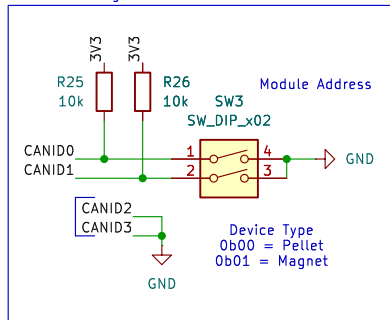
External clock



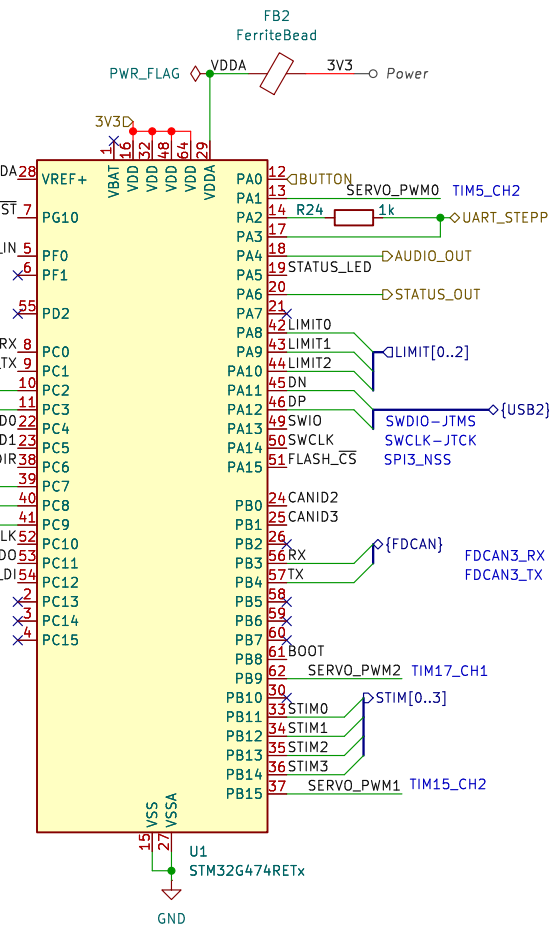
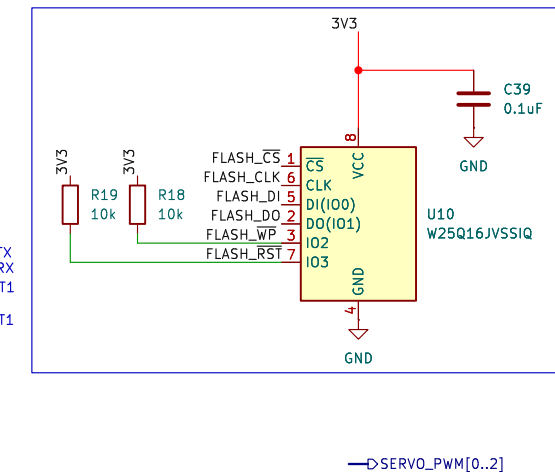
Decoupling Capacitor Central



CAN Addressing



SPI Flash



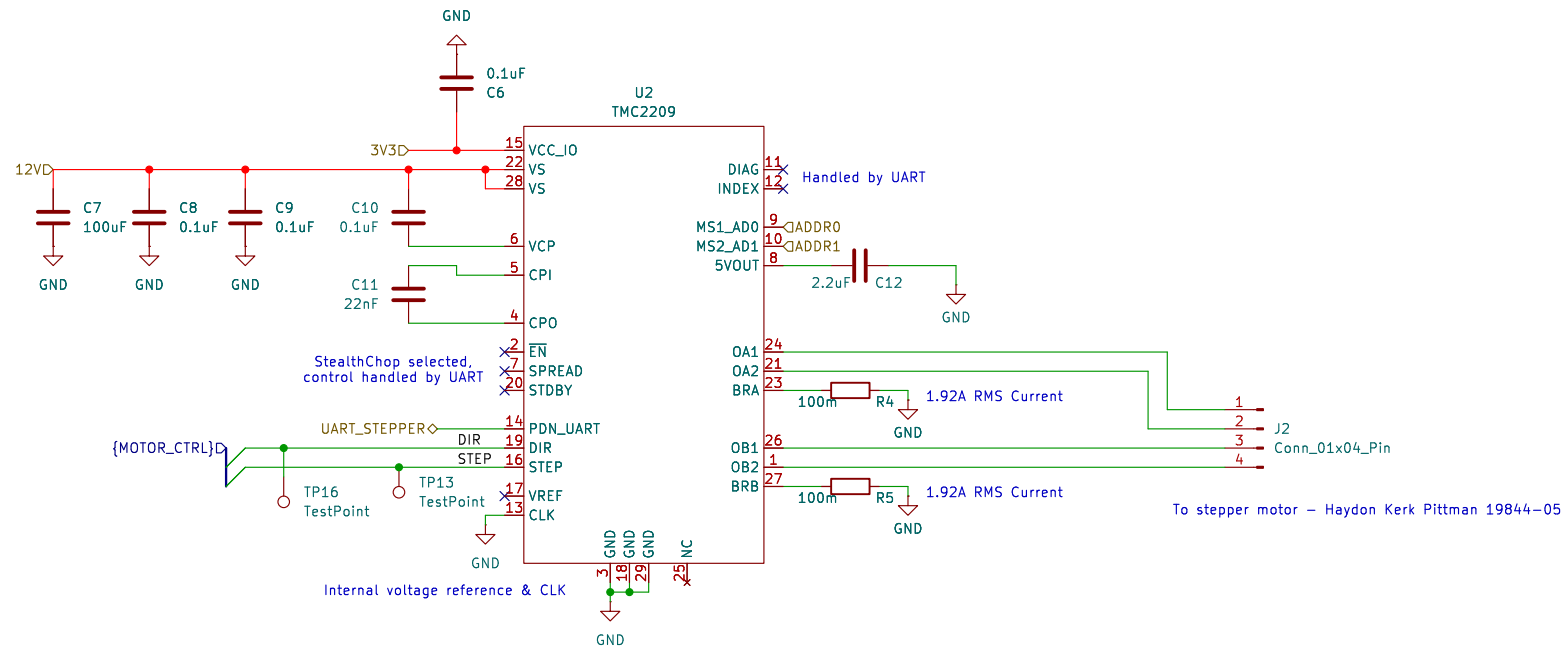
Prepared by LeafLabs, LLC
Christie Lab, CU-Anschutz

Sheet: /MCU/
File: mcu.kicad_sch

Title: Autotrainer Pellet Module – MCU

Size: A4 Date: 2024-08-12
KiCad E.D.A. 8.0.4

Rev: v1.0
Id: 2/10



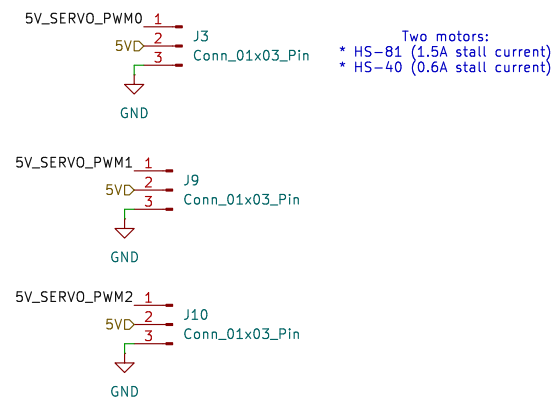
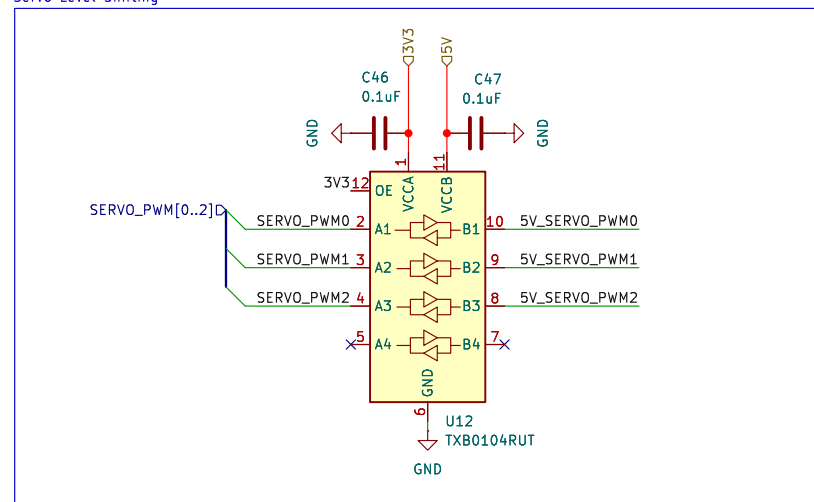
Prepared by LeafLabs, LLC
Christie Lab, CU-Anschutz

Sheet: /Stepper Driver/
 File: stepper.kicad_sch

Title: Autotrainer Pellet Module – Stepper

Size: A4	Date: 2024-08-12	Rev: v1.0
KiCad E.D.A. 8.0.4		Id: 3/10

Servo Level Shifting



Prepared by LeafLabs, LLC

Christie Lab, CU-Anschutz

Sheet: /Servo Driver/

File: servo.kicad_sch

Title: Autotrainer Pellet Module – Servo

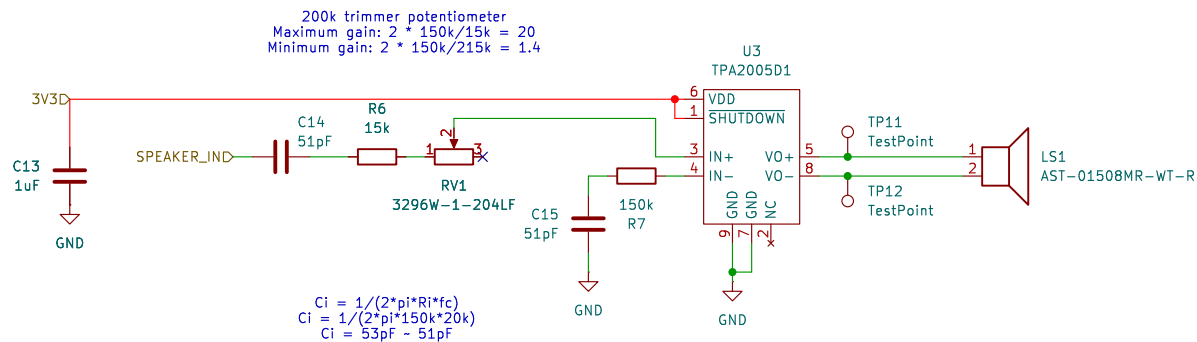
Size: A4

Date: 2024-08-12

Rev: v1.0

KiCad E.D.A. 8.0.4

Id: 4/10



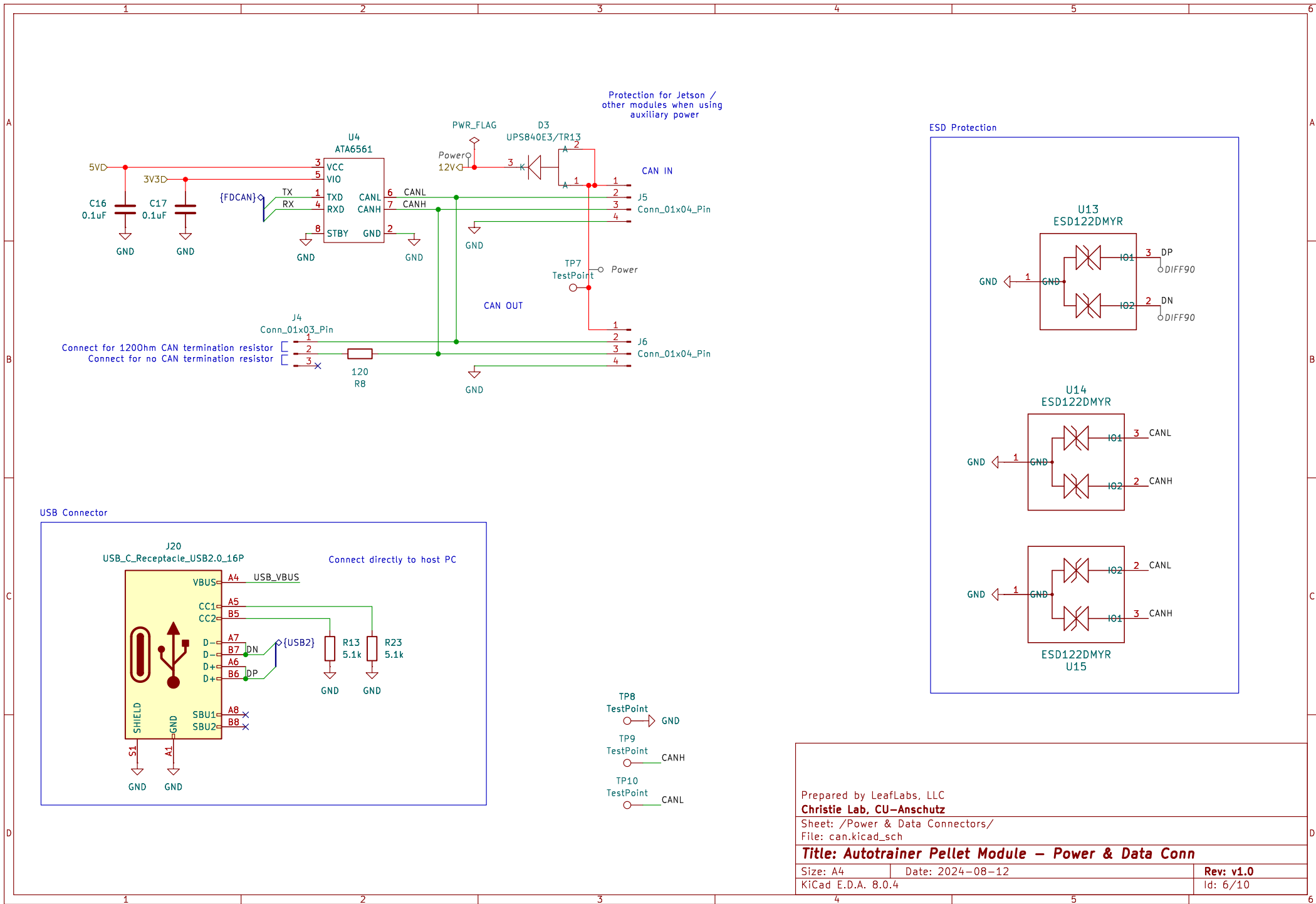
Mouse audible range is 1k-91kHz
Human audible range is 20Hz-20kHz
Design for audio between 1k-20kHz

Prepared by LeafLabs, LLC
Christie Lab, CU-Anschutz

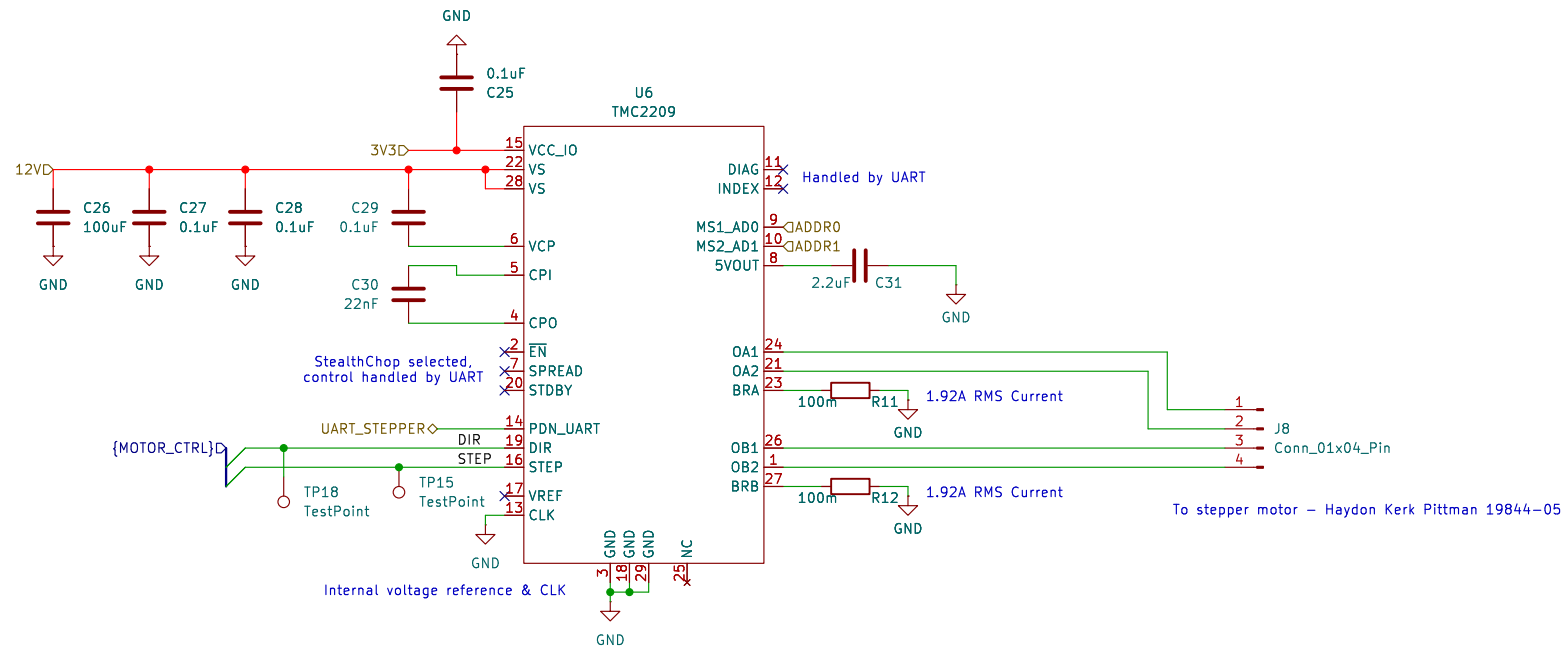
Sheet: /Speaker/
File: speaker.kicad_sch

Title: Autotrainer Pellet Module – Speaker

Size: A4	Date: 2024-08-12	Rev: v1.0
KiCad E.D.A. 8.0.4		Id: 5/10







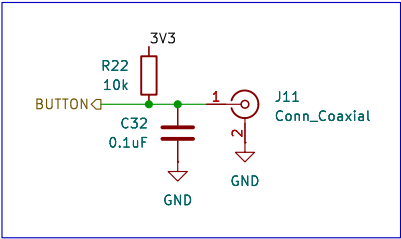
Prepared by LeafLabs, LLC
Christie Lab, CU-Anschutz

Sheet: /Stepper Driver2/
 File: stepper.kicad_sch

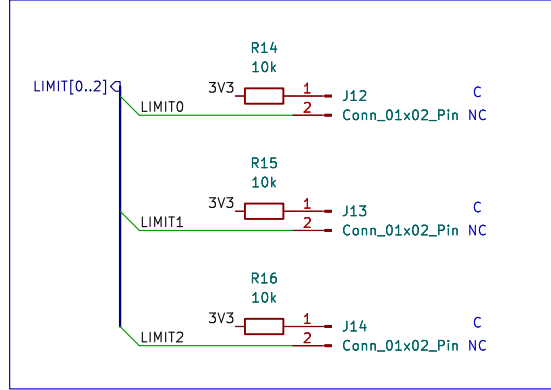
Title: Autotrainer Pellet Module – Stepper

Size: A4	Date: 2024-08-12	Rev: v1.0
KiCad E.D.A. 8.0.4		Id: 8/10

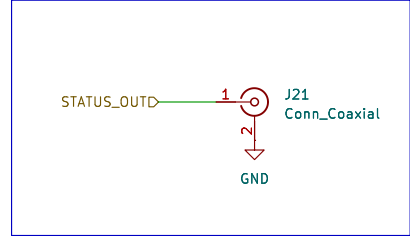
Connection to external button



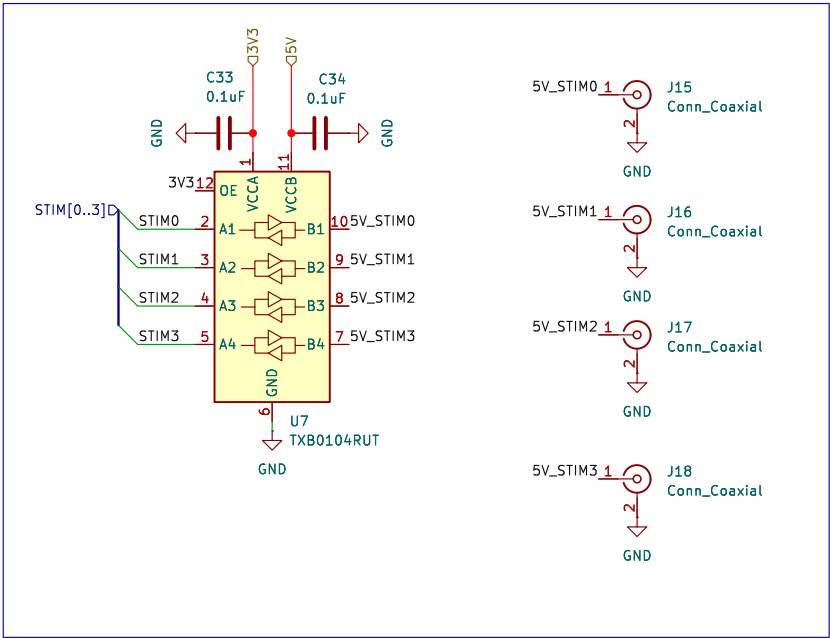
Connections to Limit Switches



Analog status indicator



External Equipment



Prepared by LeafLabs, LLC
Christie Lab, CU-Anschutz

Sheet: /External IO/
File: stim.kicad_sch

Title: Autotrainer Pellet Module – External IO

Size: A4

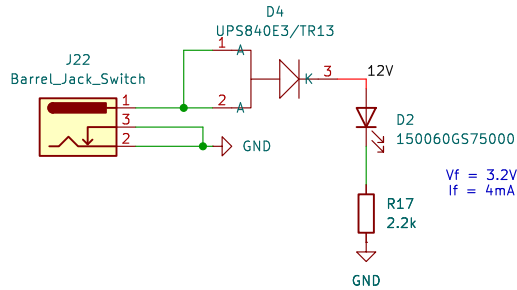
Date: 2024-08-12

Rev: v1.0

KiCad E.D.A. 8.0.4

Id: 11/10

Aux Power In



12V to 5V converter
Max expected current consumption: 4.2 A

Inductor selection math

$$I_{pp} = V_{out}/V_{in(max)} * (V_{in(max)} - V_{out})/(L_o * f_{sw})$$

$$I_{pp} = 5/12.1 * (12.1 - 5)/(3.3\mu H * 550 \text{ kHz})$$

$$I_{pp} = 1.6 \text{ A}$$

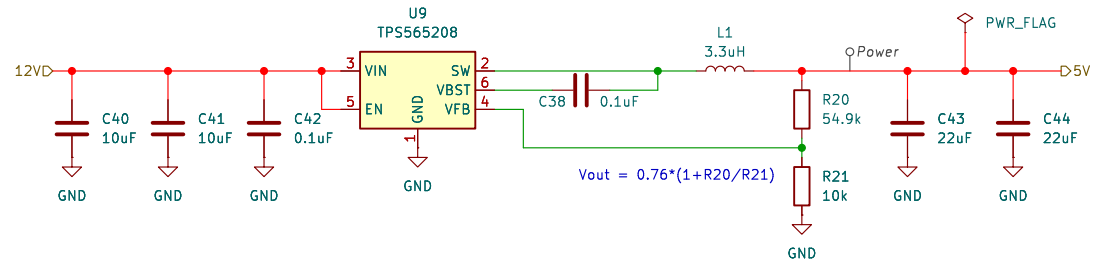
$$I_{peak} = I_o + I_{pp}/2 = 4.2 + 1.6 = 5.8 \text{ A}$$

$$I_o(rms) = \sqrt{I_o^2 + 1/12 I_{pp}^2} = 4.23 \text{ A}$$

Capacitor current

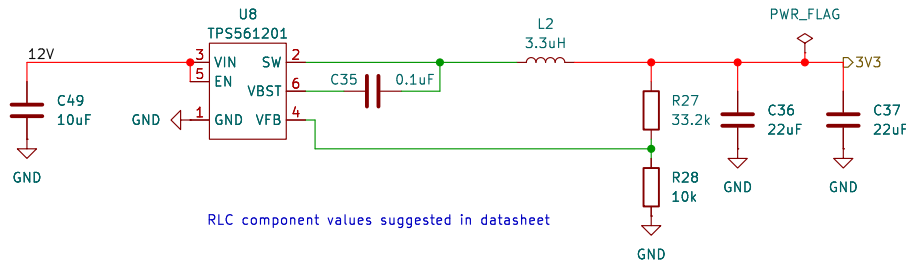
$$I_{co(rms)} = (V_{out} * (V_{in} - V_{out})) / (\sqrt{12} * V_{in} * L * f_{sw})$$

$$I_{co(rms)} = 1.11 \text{ A}$$



12V to 3V3 Converter

Expected current consumption: 126mA
Supports max 500mA



RLC component values suggested in datasheet

$$I_{pp} = 3.3/12.1 * (12.1 - 3.3)/(3.3\mu H * 580\text{kHz}) = 1.25 \text{ A}$$

$$I_{peak} = 0.5 + 1.25/2 = 1.125 \text{ A}$$

$$I_{to(rms)} = \sqrt{0.5^2 + 1/12 I_{pp}^2} = 0.62 \text{ A}$$

TP1
TestPoint
12V

TP2
TestPoint
5V

TP3
TestPoint
3V3

TP4
TestPoint
GND

TP5
TestPoint
GND

TP6
TestPoint
GND

Prepared by LeafLabs, LLC
Christie Lab, CU-Anschutz

Sheet: /Power/
File: power.kicad_sch

Title: Autotrainer Pellet Module – Power

Size: A4
Date: 2024-08-12
KiCad E.D.A. 8.0.4

Rev: v1.0
Id: 12/10